Port City International University

Excellence in Higher Education

Flood-Guard

Predicts Flood Possibilities in Bangladesh

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Introduction & Motivation

Context

Bangladesh is a climate-vulnerable nation facing annual flooding disasters.

Problem

Traditional forecasting often fails to capture non-linear climate shifts.

Motivation

We need an automated, data-driven system to predict *exact* rainfall amounts for early disaster preparedness.

Goal

Bridge the gap between historical patterns and modern anomalies using ML.

Problem Statement

Current Issue

Existing flood warnings rely on static thresholds or expensive physical simulations.

The Gap

Few systems use a "Consensus Approach" combining traditional ML stability with Deep Learning sensitivity.

The Challenge

Accurately forecasting monthly rainfall and temperature using limited historical features (Month/Year) from 1901–2020.

Objectives

01

Primary

Develop a model to forecast Rainfall (mm) and Temperature (°C) with >80% accuracy.

02

Secondary

Implement and compare Random Forest (ML) vs. Artificial Neural Networks (DL).

03

Final Goal

Create a "Flood Risk Alert" system triggering at the 300mm danger threshold.

Literature Review

No	Title	Author	Publisher & Year	Key Contribution	Dataset
1	Prediction of Rainfall in Bangladesh	Mahfuz Ahmed Azmain et al.	IEEE (2022)	Random Forest achieved strong baseline performance with 91% accuracy for regional rainfall prediction.	Bangladesh Regional Rainfall Data
2	Flood Prediction in Bangladesh Using Machine Learning and Explainable AI	Refat Ishrak Hemel, Sadman Sakib	IEEE (2024)	Demonstrated the importance of explainable models to interpret flood prediction decisions.	Meteorological & Hydrological Data
3	Relations of Water-logging with Heavy Rainfall in Dhaka City (1991-2021)	Mohammad Nazrul Islam	JETIR (2022)	Identified 300 mm monthly rainfall as a critical threshold causing urban flooding.	Dhaka Urban Rainfall Data

Methodology Overview

Step 1

Data Collection (BMD Archives 1901-2020).

Step 2

Preprocessing (Scaling & Cleaning).

Step 3

Data Splitting for Training and Testing

Step 4

Model Training (Model A: Random Forest, Model B: ANN).

Step 5

Model Evaluation

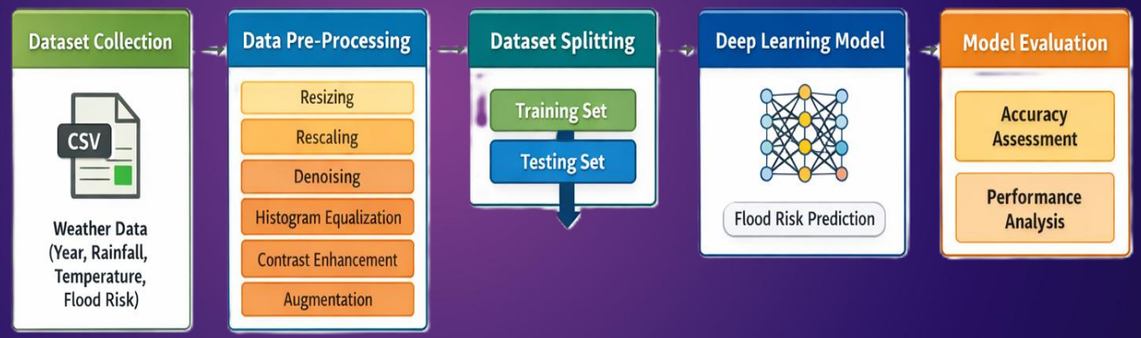


Fig 1: Architecture

The Dataset

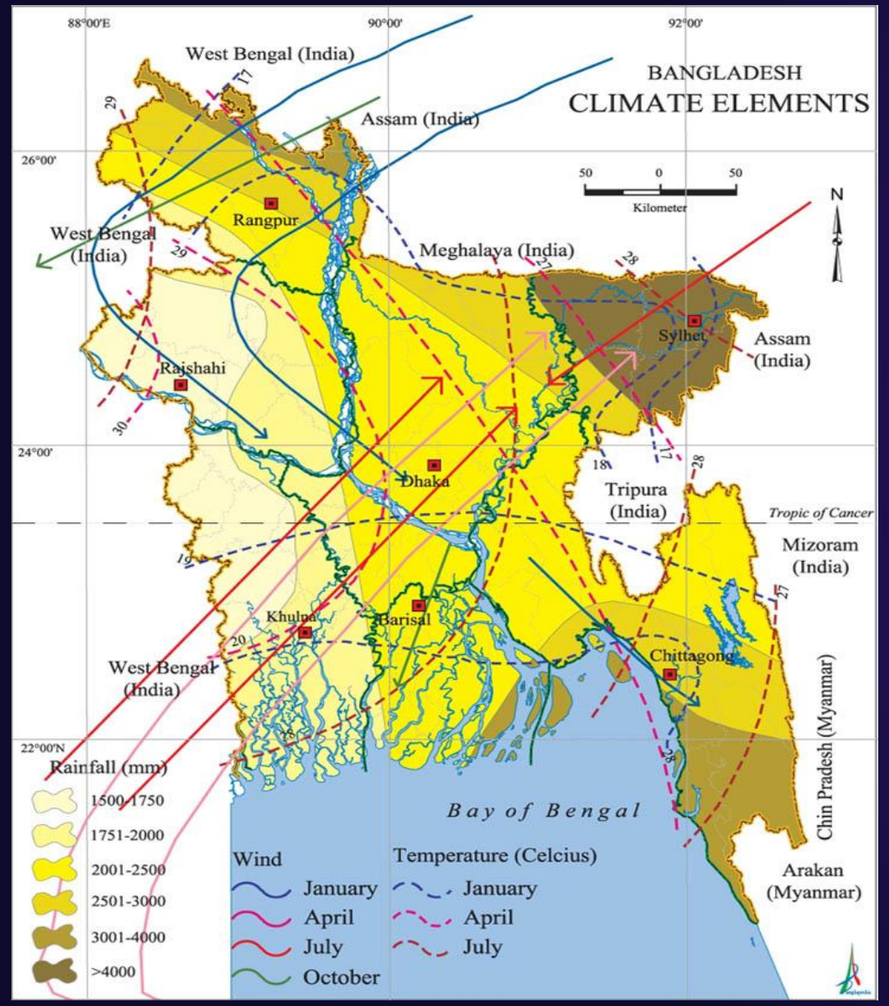


Fig 2: Bangladesh Weather Map

- **Source:** Bangladesh Meteorological Department (BMD) Archives, through Kaggle.
- **Volume:** ~1,474 Monthly Records (1901–2020).
- **Inputs (Features):** Month (Seasonality), Year (Long-term trends).
- **Outputs (Targets):** Rainfall (mm), Temperature (°C).

System Requirements



Hardware

AMD Ryzen 5 3500, 16GB RAM



Software

Python 3.10+, Google Colab Environment.



Libraries

Scikit-Learn (ML), TensorFlow/Keras (Deep Learning), Pandas, Matplotlib.

Data Pre-Processing

Cleaning

Renamed columns (tem -> Temperature, rain -> Rainfall). Verified zero null values.

Scaling

Applied StandardScaler to normalize inputs (Crucial for Neural Network performance).

Splitting

80% Training Data, 20% Testing Data.

Result & Discussion

Accuracy Results

Metric:

R^2 Score (Coefficient of Determination).

Random Forest:

~76% Accuracy (Rainfall), ~96% (Temp).

Neural Network:

~82% Accuracy (Rainfall), ~96% (Temp).

Verdict:

Both models are reliable, with ANN showing a slight edge in capturing rainfall complexity.

Result Summary

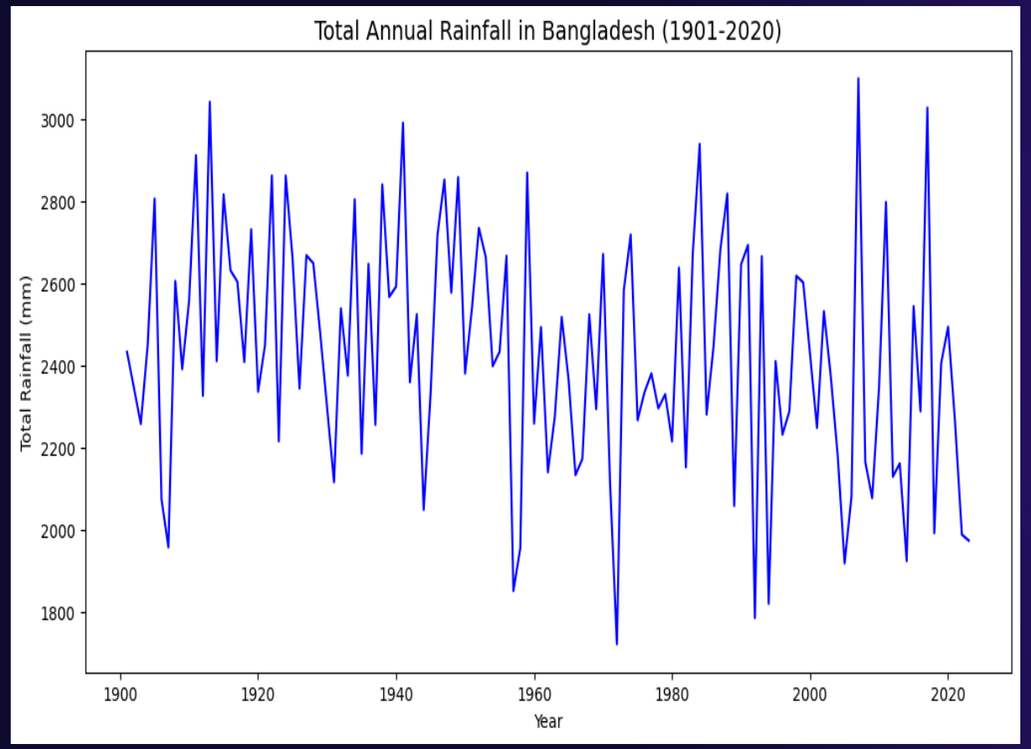


Fig 3: Total Rainfall in Bangladesh (1901-2020)

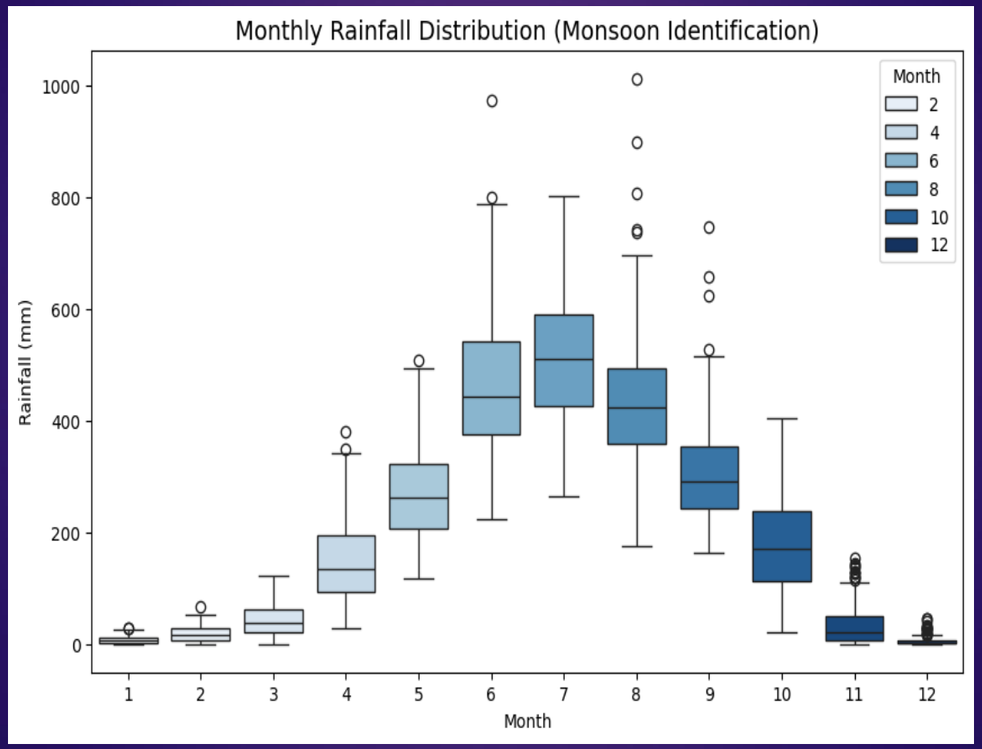


Fig 4: Monthly Rain Distribution

Result Summary

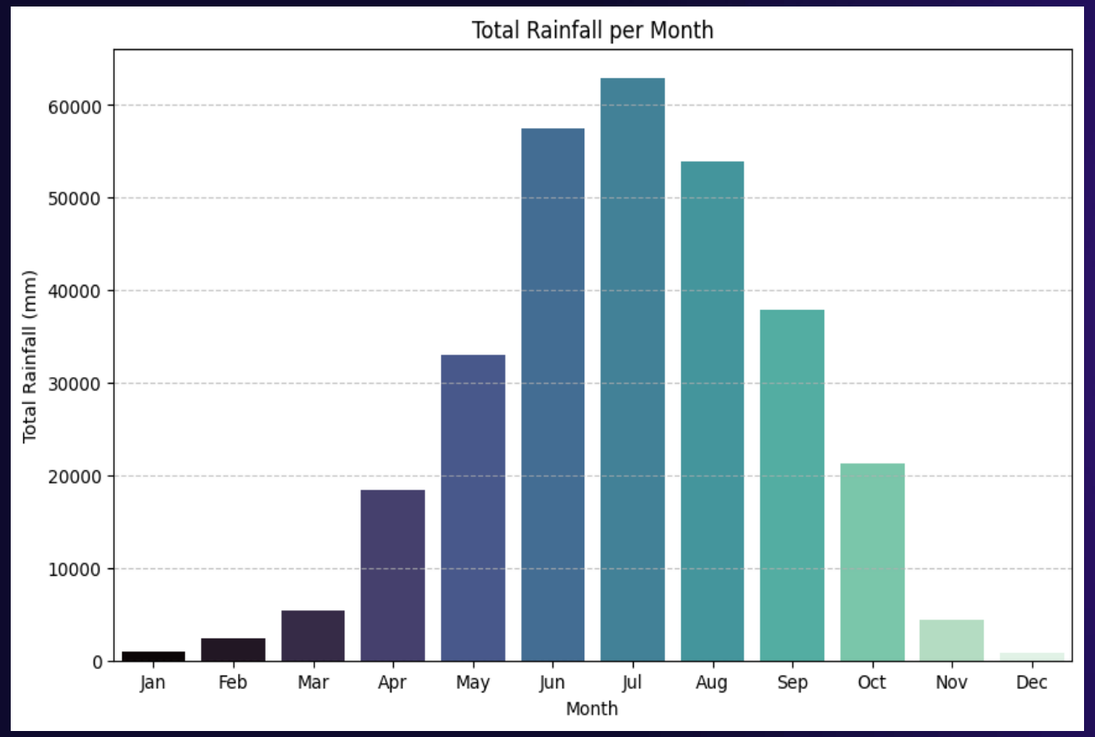


Fig 5: Total Rainfall per Month in Bangladesh

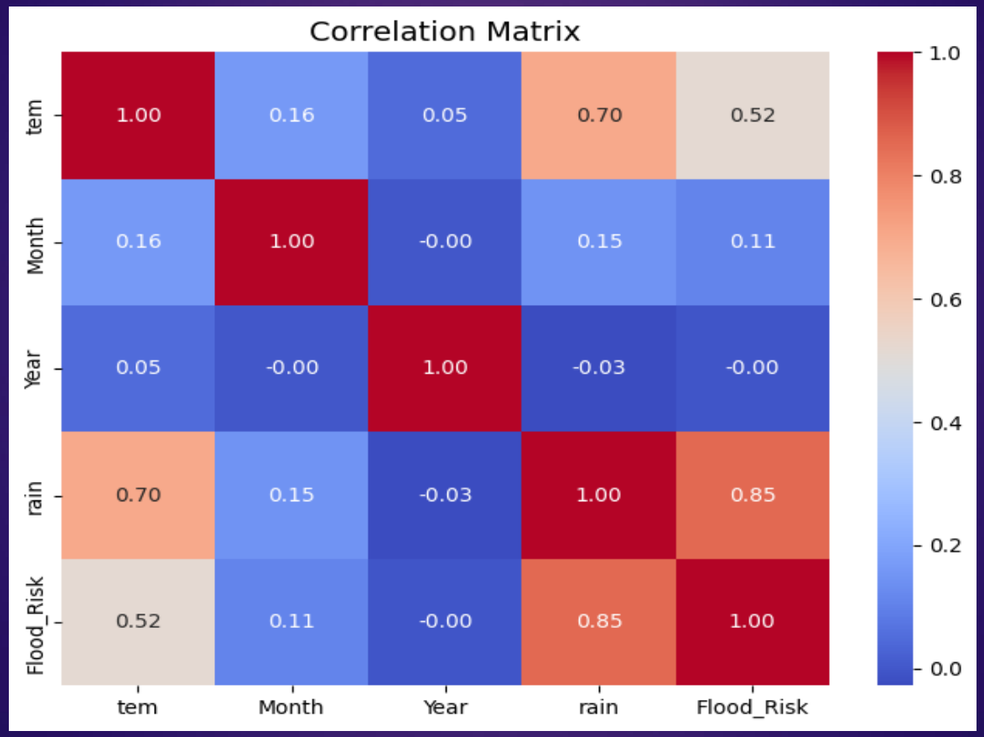


Fig 6: Correlation Matrix

Result Summary

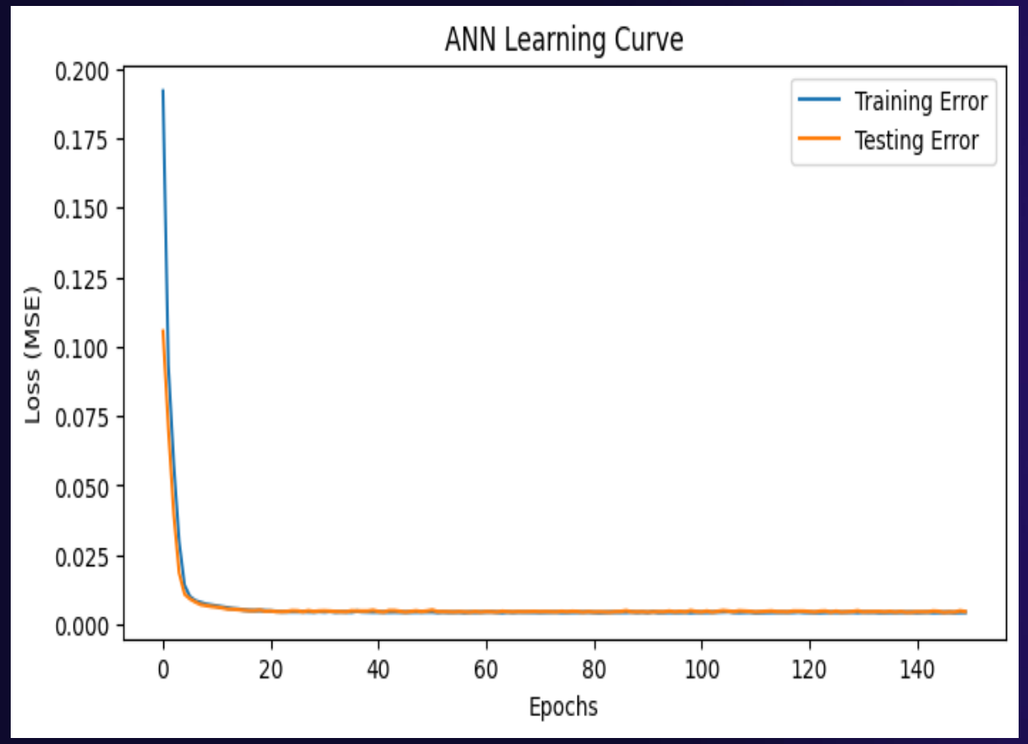


Fig 7: ANN Leaning Curve

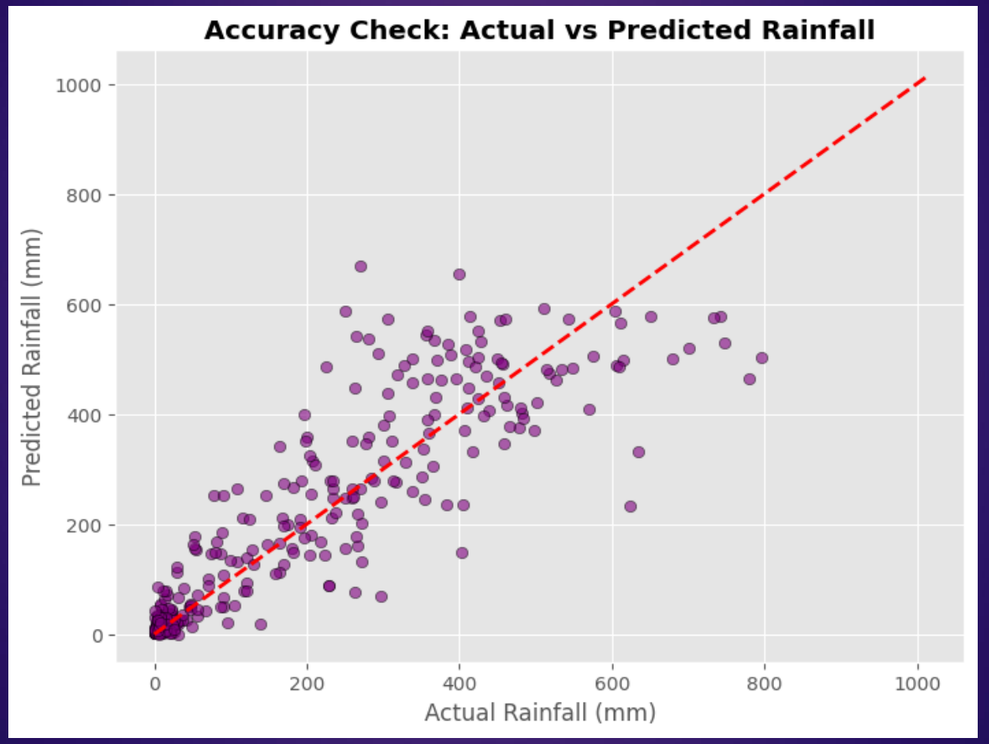


Fig 8: Accuracy Check

Result Summary

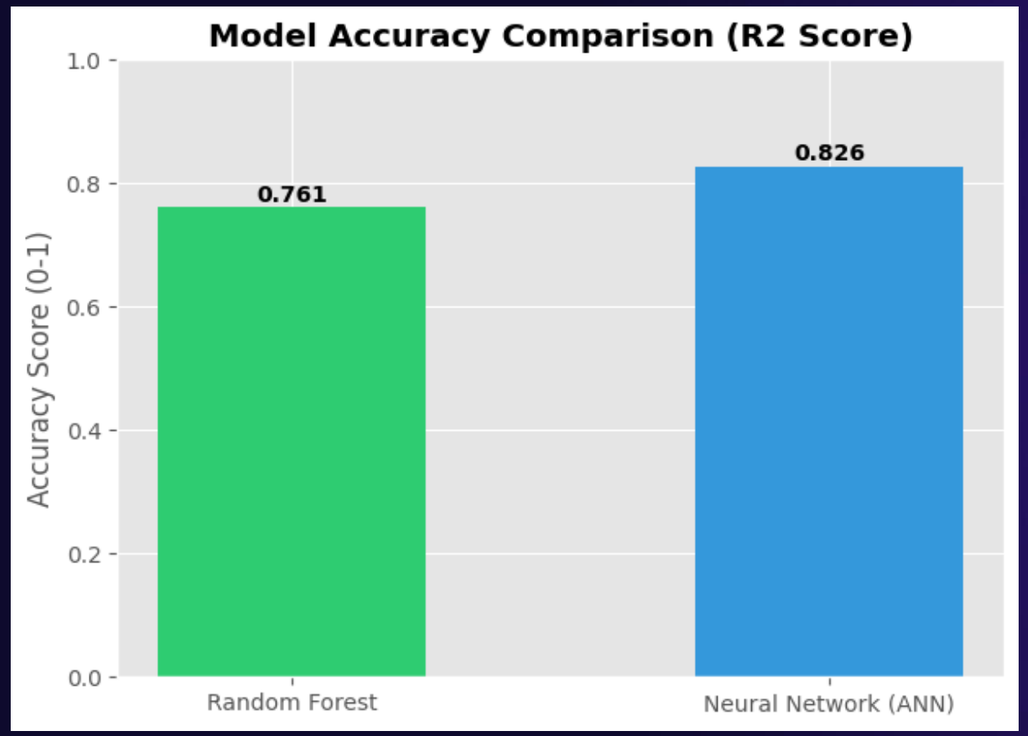


Fig 9: Accuracy Comparison (RF vs ANN)

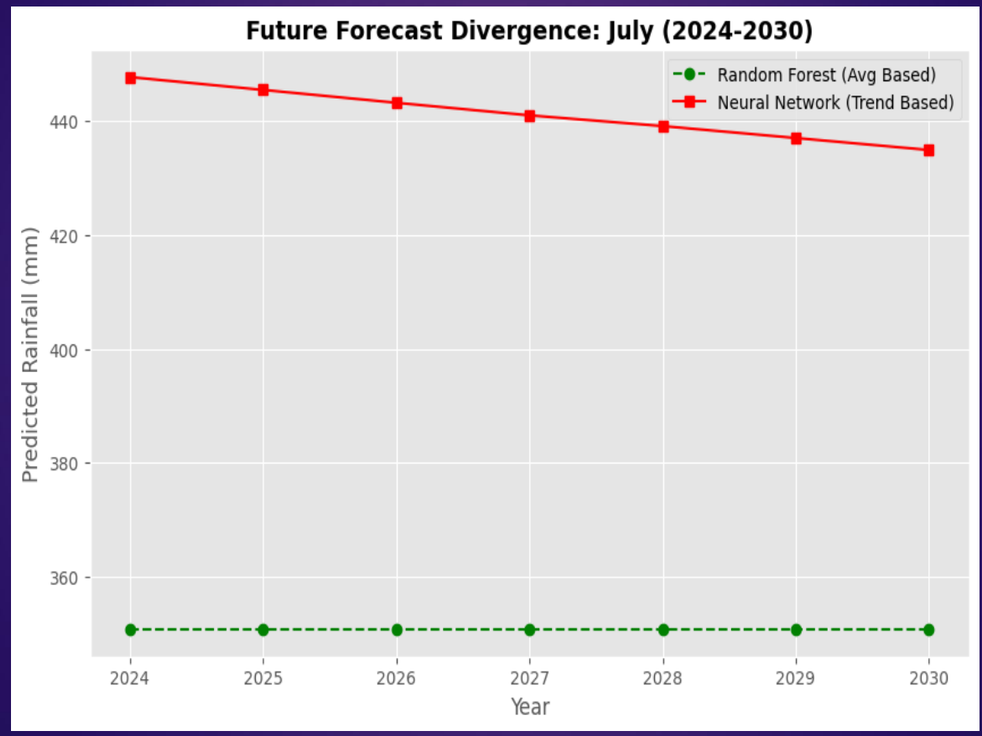


Fig 10: Future Forecast Divergance

Comparative Analysis

Divergence:

For July 2031, RF predicted 350.85 mm, while ANN predicted 410.07 mm.

Interpretation:

RF relies on 100-year averages. ANN detected a recent drying trend in the specific month.

We present *both* numbers to offer a "Best Case" vs. "Worst Case" scenario to policymakers.

Conclusion

Success:

Developed "Flood-Guard," a functional dual-engine AI system.

Impact:

Proves Machine Learning is viable for long-term forecasting in Bangladesh.

Innovation:

Mitigated individual model weaknesses by combining ML and DL architectures.

References

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